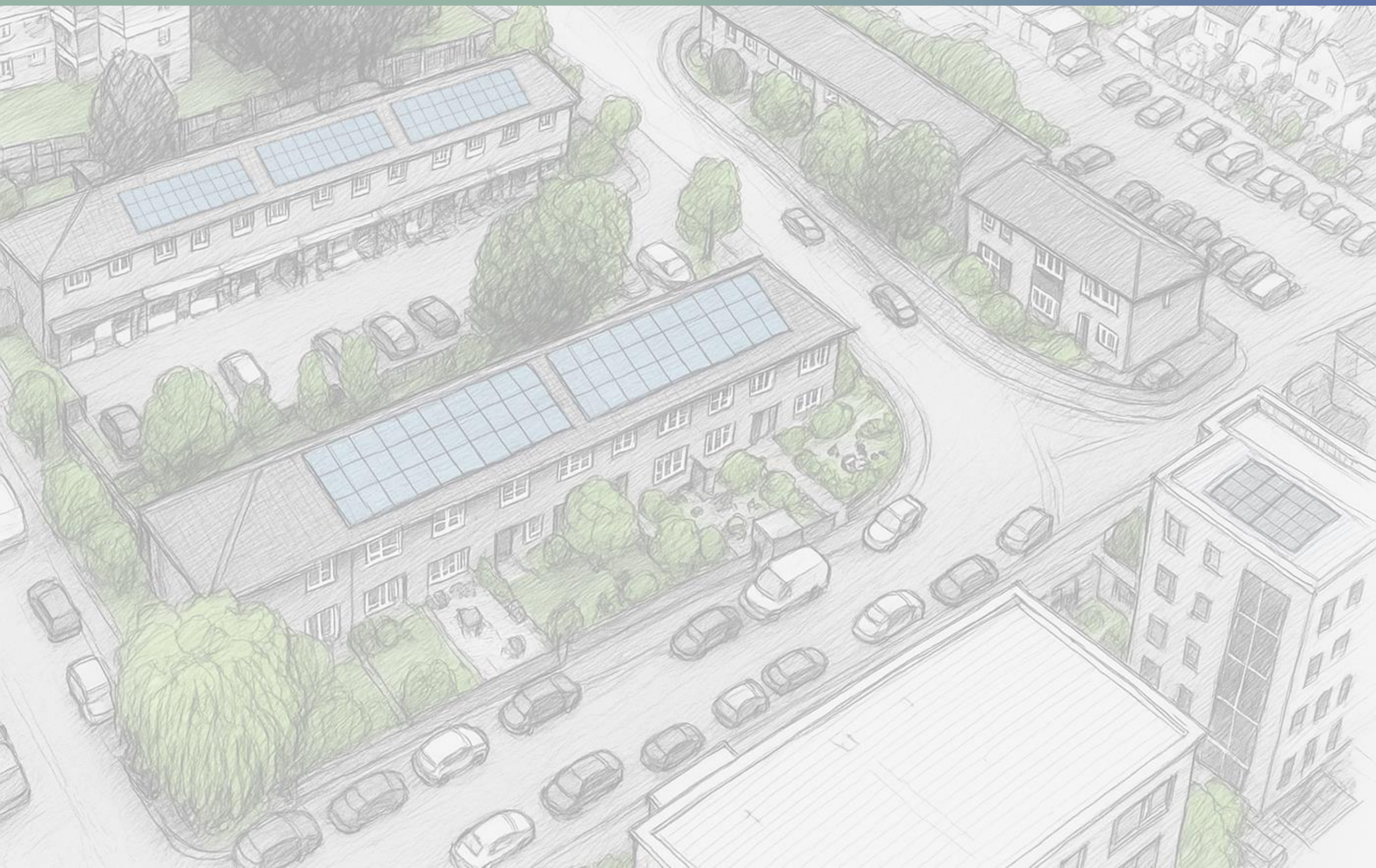




Executive Summary

Using our in-house asset intelligence tool PowerAsset, Energy Specifics Group has reviewed over **6,000** St Albans homes to identify retrofit opportunities across the housing portfolio. Our analysis has identified **1,115** homes currently below EPC C, with the potential for further properties to be identified from the **642** homes without a lodged EPC. These homes represent a significant opportunity to improve energy efficiency, reduce carbon emissions, lower residents' energy bills, and support St Albans City & District Council's decarbonisation ambitions.

Our assessment combines EPC data, damp and mould, climate projections, overheating risk, IMD, GIS mapping, planned works, component replacement programmes, and other asset intelligence. Through bespoke tools including fabric and visual archotyping, geographical clustering, property level overheating scoring, and investment optimisation, PowerAsset identifies the right properties, at the right time, ensuring retrofit programmes are aligned with wider investment plans to maximise value for money, minimise resident disruption, and improve delivery efficiency.



Aerial View of St Albans

Purpose of the Report

This report supports St Albans City & District Council in optimising its retrofit programmes through enhanced, data driven analysis of its housing stock. Using PowerAsset, Energy Specifics Group has assessed the wider portfolio to identify priority retrofit opportunities, high impact property archetypes, deliverable geographical clusters, and climate related risks, including overheating. By combining EPC, GIS, climate, IMD and asset datasets with bespoke archotyping, clustering and overheating scoring tools, the report provides a strategic evidence base to maximise delivery efficiency, align retrofit with planned investment, and achieve the greatest value for money.

- **Solid Wall Properties**

St Albans City & District Council owns approximately 581 solid wall properties, with 179 currently below EPC C, including around 432 houses, 138 flats, and a small number of maisonettes and bungalows. These represent a significant opportunity for fabric-first retrofit. The largest concentration is in St Albans city itself. Using PowerAsset's clustering and archotyping capabilities, similar properties can be grouped to reduce design variation, repeat surveys, and logistics complexity while supporting a scalable retrofit programme.

- **Non-Trad Properties**

St Albans City & District Council owns approximately 995 system built and timber frame properties, of which 319 are currently below EPC C. Although this represents a relatively small proportion of the overall stock, the properties are concentrated across Wheathampstead, London Colney and parts of the Sandridge area. This concentration presents an opportunity to develop standardised retrofit solutions for similar construction types, reducing design complexity, survey requirements and procurement costs. In addition, these homes are well suited to a dedicated delivery programme, enabling specialist supply chains, repeatable installation methods, and consistent technical solutions that improve programme efficiency, quality and value for money.

- **Flats / Blocks**

St Albans City & District Council owns approximately 2,532 flats, of which 671 are currently below EPC C. Around two thirds are cavity wall construction and one third are system built or solid wall, presenting a significant opportunity for targeted fabric first improvements. The stock is concentrated across Wheathampstead and London Colney, creating opportunities for coordinated, block based retrofit programmes aligned with planned works and component replacement. PowerAsset's visual archotyping and data cleansing can also identify properties incorrectly classified as flats, such as maisonettes, improving stock accuracy and retrofit planning.

- **Cavity Wall Properties (CWI)**

St Albans City & District Council owns approximately 3,836 cavity wall properties, with 592 currently below EPC C. A small proportion are off the gas network, creating opportunities to combine cavity wall improvements with low-carbon heating and renewable technologies. These homes are spread across the district in clusters, enabling area-based delivery approaches that reduce mobilisation costs and maximise economies of scale.

FIG 1: Sample visualisation of a Client's Asset data for integrating planned maintenance with planned decarbonisation, including accurate budget forecasting

Fig2: Sample Analysis of a Client's Damp & Mould Data

ESG's Role and Strategic Support

ESG helps landlords create and deliver investment programmes, including retrofit, that are efficient, scalable and strategically aligned. We support clients throughout the programme lifecycle, from stock analysis and investment planning to design, funding, procurement support and delivery.

Retrofit Consultancy, Design and Delivery Planning: ESG provides end to end retrofit consultancy, delivering PAS 2035 compliant designs tailored to property archetypes, client objectives and site constraints. Using BIM, digital twins and building performance modelling, we develop coordinated, future ready solutions that reduce delivery risk and support long term asset investment.

Decarbonisation and Investment Strategy: We develop whole stock decarbonisation strategies, identifying priority properties and aligning retrofit with planned maintenance, component replacement, funding opportunities and wider investment programmes to maximise value for money.

Funding Identification: ESG identifies properties best suited to self-funded retrofit or external funding, including SHDF and other government programmes, ensuring investment is targeted where it delivers the greatest impact.

Neighbourhood and Archetype Targeting: Using geographical clustering and bespoke archotyping, ESG identifies repeatable delivery opportunities that reduce design variation, improve logistics, lower programme costs and support area-based delivery.

Stock Intelligence and Risk Assessment: ESG reviews and refines stock data by identifying data quality issues, leasehold constraints, planning considerations, heritage assets, and properties at risk of damp, mould and overheating using asset, environmental and climate datasets. This enables targeted interventions and more informed design decisions while ensuring programmes are technically robust and deliverable (see Figs. 1 & 2).

In summary, ESG combines advanced data analytics, retrofit expertise and delivery planning to help housing associations develop investment programmes that are practical, fundable and future ready.

Portfolio Insights

PowerAsset has analysed **6,193** St Albans properties, identifying **1,115 homes below EPC C**, including **1,048 EPC D**, **61 EPC E**, and **5 EPC F** properties. These homes represent the highest priority for future retrofit investment and the greatest opportunity to improve energy performance across the portfolio.

The housing stock is predominantly made up of **2,388 houses**, **2,532 flats**, **340 bungalows** and **246 maisonettes**. In terms of construction, the portfolio contains **3,836 cavity wall**, **581 solid wall**, **995 non-traditional construction (72 timber frame and 923 system built)** enabling retrofit programmes to be developed around repeatable construction archetypes and standardised technical solutions. The stock is also primarily made up of **semi-detached (2,267)**, **mid-terrace (1,404)** and **end-terrace (1,140)** homes, providing further opportunities for efficient archetype-based delivery.

PowerAsset has identified **642 properties without a lodged EPC**, **336 properties with expired EPCs**, and **143 EPCs approaching expiry**. These properties present both a risk and an opportunity. Homes with expired EPCs may have undergone energy efficiency improvements in recent years that are not reflected within the EPC Register, meaning some may already achieve EPC C or above. Refreshing EPC data and modelling homes without EPCs will improve stock intelligence, provide a more accurate view of portfolio performance, and support better informed investment decisions.

The analysis also identified about **175 off gas properties**, presenting an opportunity to prioritise low carbon heating technologies alongside fabric first improvements. In addition, **27 listed buildings** and **752 properties within conservation areas** have been identified, enabling planning constraints to be considered early and ensuring appropriate retrofit solutions are developed from the outset.

By combining EPC, asset, planning and geographical datasets with bespoke archotyping and clustering, PowerAsset enables St Albans City & District Council to prioritise investment where it will deliver the greatest carbon reduction, programme efficiency and long-term value.

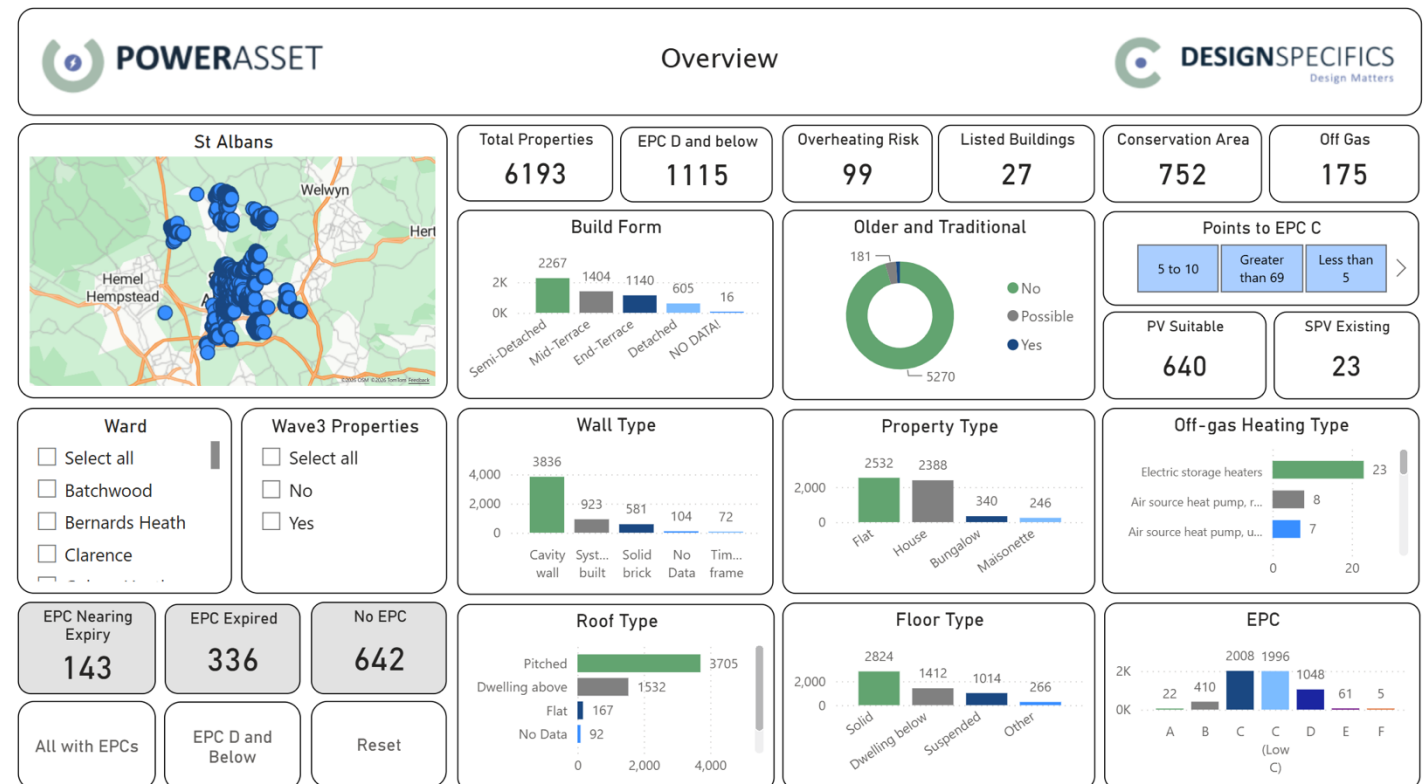


Fig 3: Analysis of St Albans City & District Council's property portfolio

Visual Validation of Archetypes

While archetype modelling based on data is foundational, further validation through visual assessment (survey imagery, satellite data, and site visits) identifies important nuances- such as extensions, orientation, or façade alterations, that influence measure suitability. These insights inform the refinement of archetype groupings and improves the precision of both measure specification and delivery sequencing.

Heritage & Planning Context

Analysis also identified **226 properties** below EPC C located within conservation areas and **13 listed buildings** likely to be subject to planning constraints, requiring tailored design solutions (e.g. internal insulation, specialist fenestration). Early-stage digital overlays supported efficient scoping and avoidance of costly rework or planning delay.

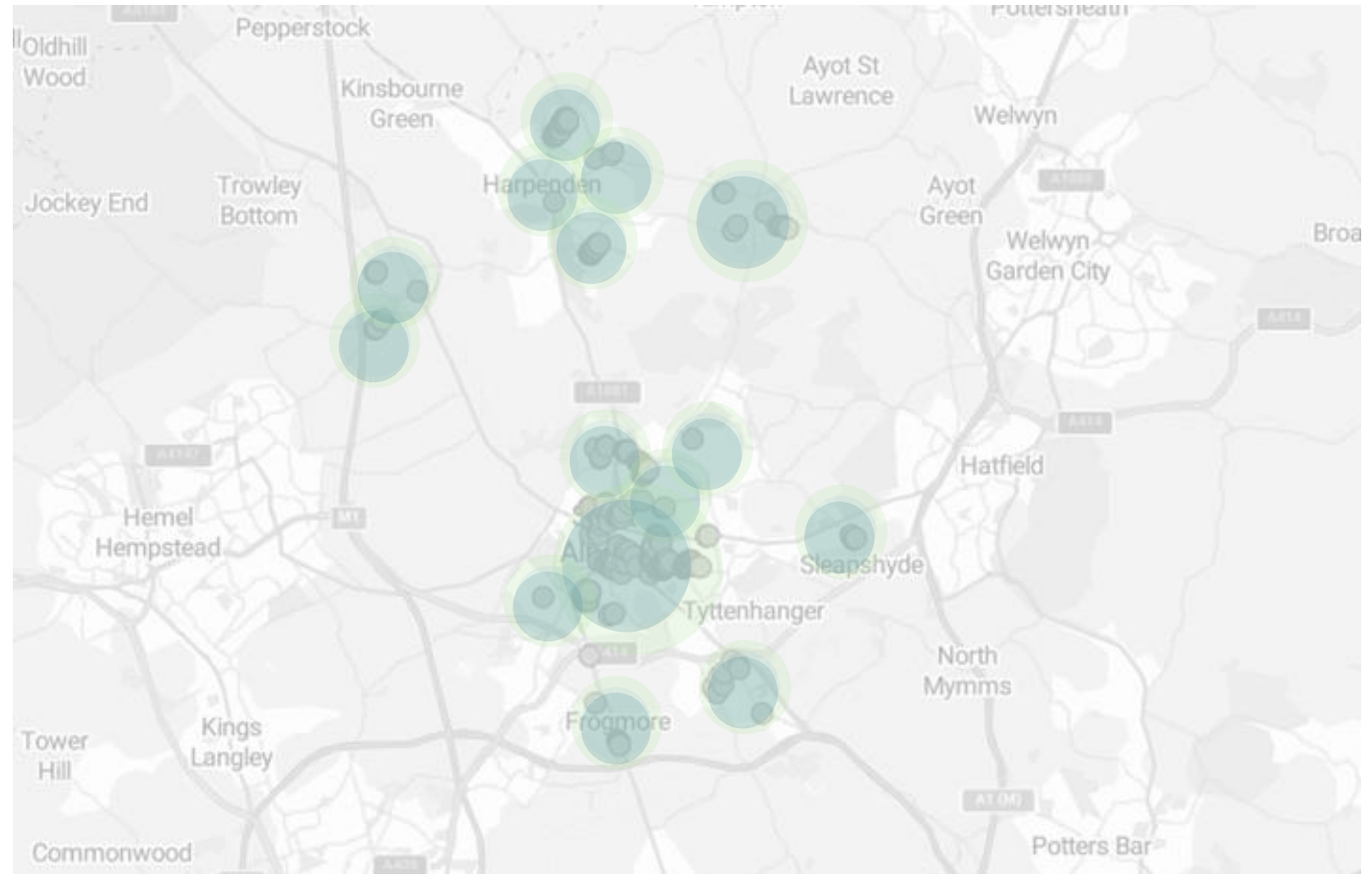
Solid Wall Property Deliverability Insights

Analysis of St Albans City & District Council housing stock shows that **179 of the 581 properties** are currently below **EPC C**, representing approximately **31% of the solid wall portfolio** and a substantial opportunity for future retrofit investment. When excluding flats, the number of suitable properties reduces to **85 homes**, providing a more focused opportunity for retrofit delivery where St Albans City & District Council has greater control over the building envelope and external works. Flats can present additional delivery challenges where the Council does not have full ownership of the block, as coordination of external alterations, access requirements, and resident engagement can introduce additional complexity.

Fabric-first measures, particularly **External Wall Insulation (EWI)**, require a detailed design and delivery process, with every external attachment, scaffold arrangement, material storage requirement, and road-space permit carefully coordinated. Prioritising geographically clustered properties will reduce mobilisation costs, simplify logistics, and improve delivery efficiency. This approach also creates an opportunity to maximise the value of scaffold mobilisation by coordinating complementary improvements, such as replacement of ageing windows and doors, where required. In addition, integrating renewable technologies such as solar PV and battery storage alongside fabric measures can enable entire streets or neighbourhoods to achieve a consistent energy performance standard, creating a more uniform, resilient, and future-ready housing portfolio across St Albans City & District Council's estates.

Recommendations

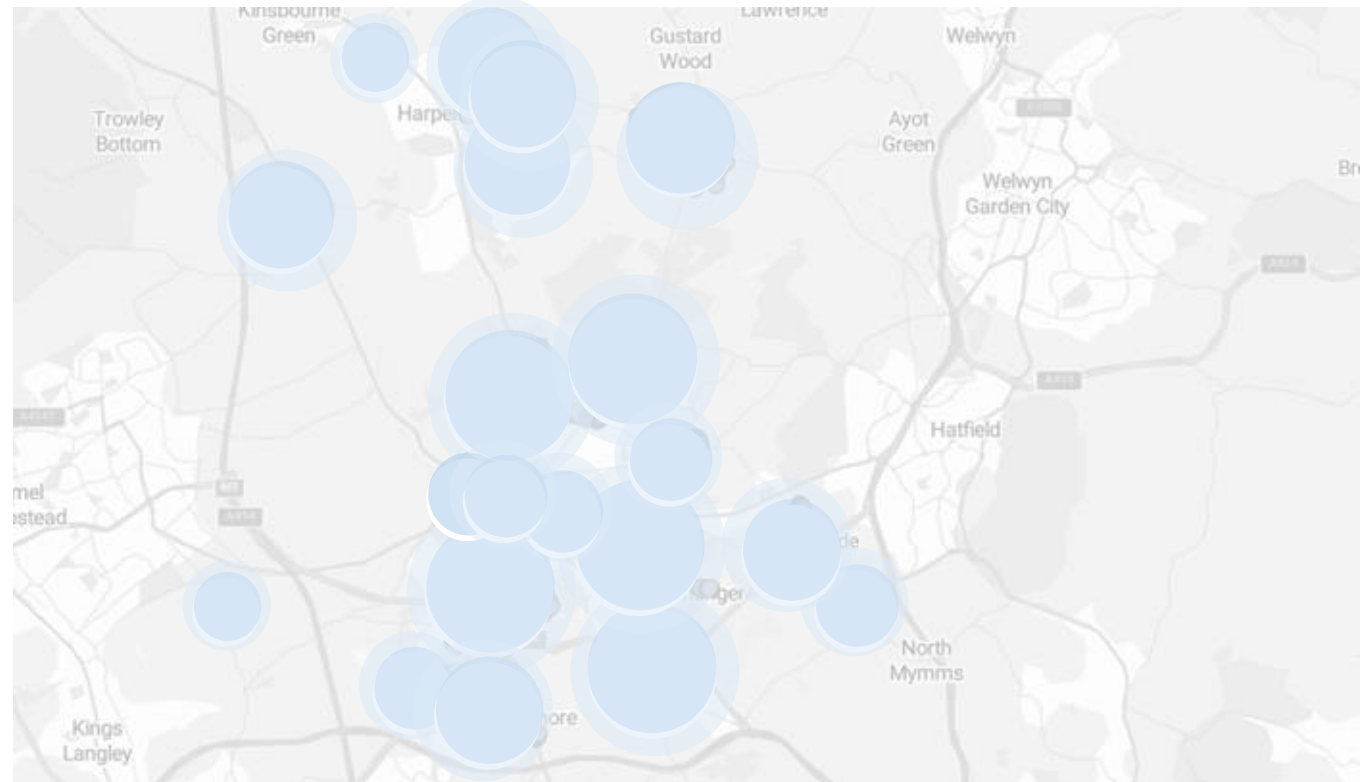
- **Prioritise clustered properties** on the same street or within defined neighbourhoods to maximise delivery efficiency, reduce scaffold and traffic management costs, and streamline contractor mobilisation.
- **Integrate fabric, renewable, and planned maintenance works** by delivering EWI alongside technologies such as solar PV, battery storage, and window/door upgrades where feasible, creating a coordinated whole-house retrofit strategy.
- **Defer isolated properties and flats where appropriate** until they can be incorporated into wider delivery programmes or where ownership constraints can be resolved, reducing standalone mobilisation costs and improving overall programme value.



Spread of Solid Wall properties across St Albans

Cavity Wall Property Deliverability Insights

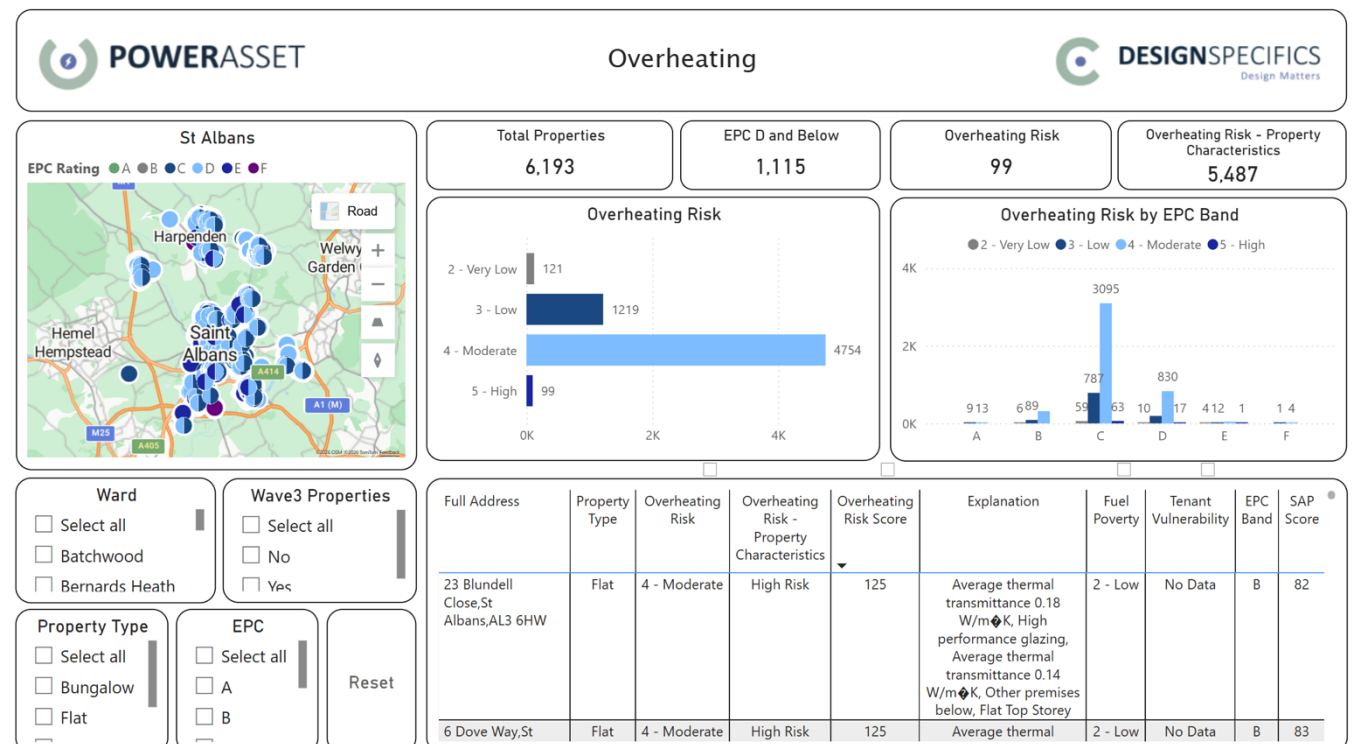
Analysis of St Albans City & District Council's cavity wall housing stock identifies **3,836 properties**, of which **592 are currently below EPC Band C**, representing a significant opportunity for large-scale retrofit delivery. When excluding flats, the number of suitable cavity wall properties reduces to **233 homes**, providing a more practical focus for measures where the Council has greater control over the whole building envelope. Flats may present additional delivery challenges where the Council does not own the entire block, as coordinating fabric improvements, external alterations, and resident access across multiple ownerships can introduce complexity.



Spread of Cavity Wall properties across St Albans City & District Council portfolio

Overheating Risk

The analysis also identifies **a significantly large number of properties subject to Overheating risk**, requiring careful consideration when assigning retrofit measures. While improving fabric performance will reduce heat loss, the combination of measures must be assessed holistically to ensure improvements do not unintentionally increase summer overheating risk. Consideration should be given to ventilation, solar gains, shading, and appropriate low-carbon technologies.



Overheating risk analysis across St Albans City & District Council portfolio

System Built Properties Deliverability Insights

Analysis of St Albans City & District Council's **system-built housing stock** identified **923 properties (with a further 72 timber frame)**, of which 319 are currently below EPC Band C. Of the 212 properties proposed for retrofit works as part of this scheme, there are 150 system-built properties. Of those 150 properties, **97 are currently below EPC Band C**.

System-built properties require careful consideration due to the risks associated with non-traditional construction methods and external cladding systems:

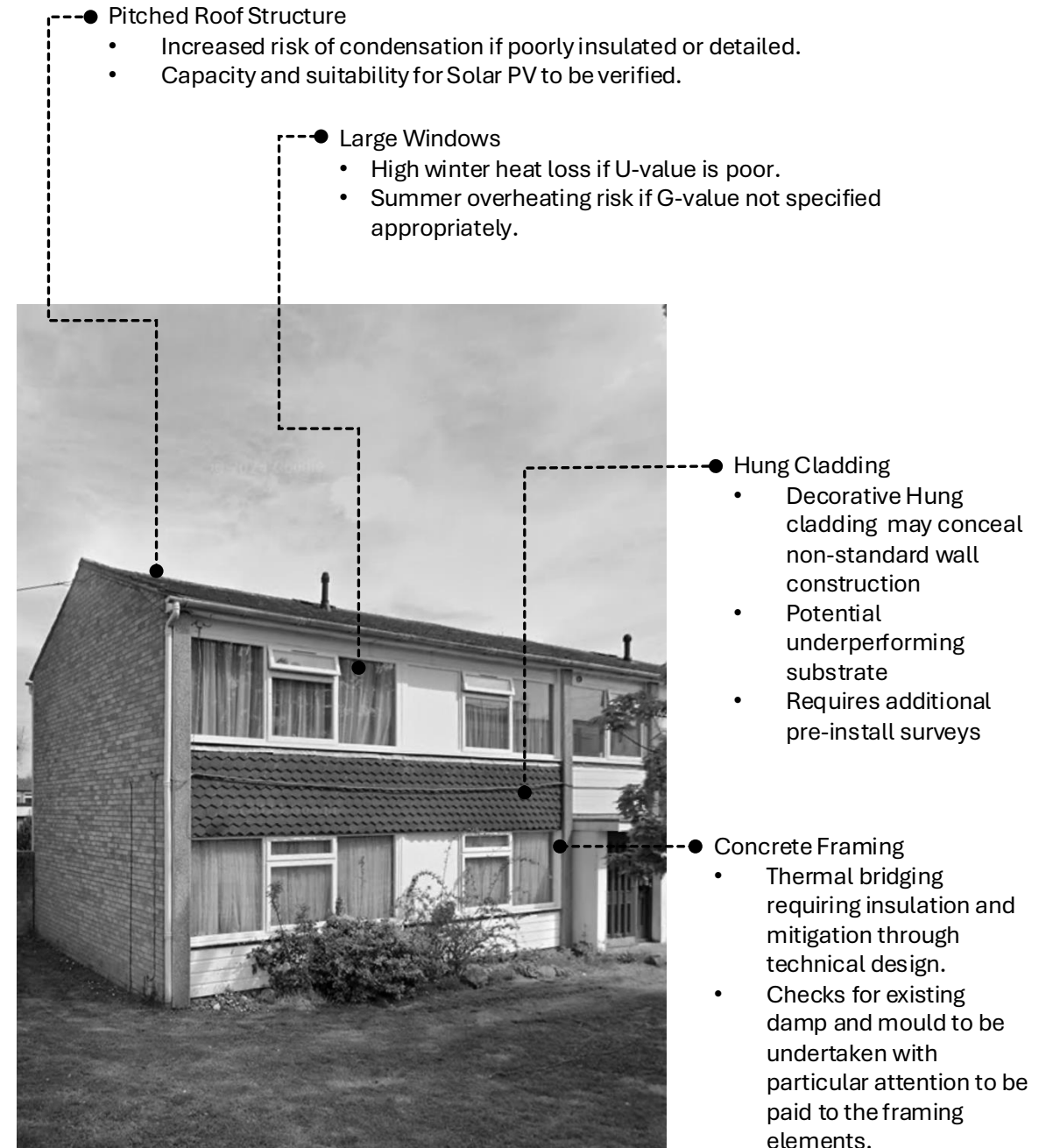
- **Cold bridging:** Non-traditional construction can contain significant thermal bridges, requiring additional detailing and mitigation measures.
- **Cladding and detailing risks:** Systems such as **hung tile cladding** require assessment before retrofit measures are confirmed to ensure that the existing construction is understood and proposed insulation can be detailed accordingly.
- **Structural uncertainty:** Elements such as timber battens, steel components, and fixings may require investigation due to potential corrosion, moisture damage, or deterioration.
- **Roof construction:** Renewable technologies such as solar PV should be prioritised where roof condition and structural capacity allow.

A review of the portfolio highlighted two particular non-traditional construction types: Gregory Industrialised and Wimpey No Fines. Of the 150 properties identified, all are flats within blocks of varying heights.

As an example, the low-rise Gregory Industrialised block type is shown on the right. This block features hung tile cladding, large windows, concrete framing, and a pitched tile roof. These construction characteristics introduce several thermal bridge junctions and may incorporate non-standard construction methods and underperforming substrates, making fabric-first retrofit more technically challenging than conventional housing. Having worked on similar archetypes, subject to additional Pre-Installation Surveys:

Recommended approach

- Prioritise loft insulation where loft is present as the initial fabric measure to improve thermal performance with minimal construction risk.
- Deploy renewable technologies, including solar PV and air source heat pumps (ASHPs), where roof condition and structural capacity are suitable, particularly for the properties currently rated EPC Band D and below.
- Replace windows and doors to reduce heat loss and present the opportunity to introduce solar gain control through the g-value specification to prevent summer overheating.
- Insulation to concrete frame and behind hung tile cladding., to reduce heat loss and mitigate thermal bridging.



Sample 'Gregory' System-Built block of flats from St Albans City & District Council Stock

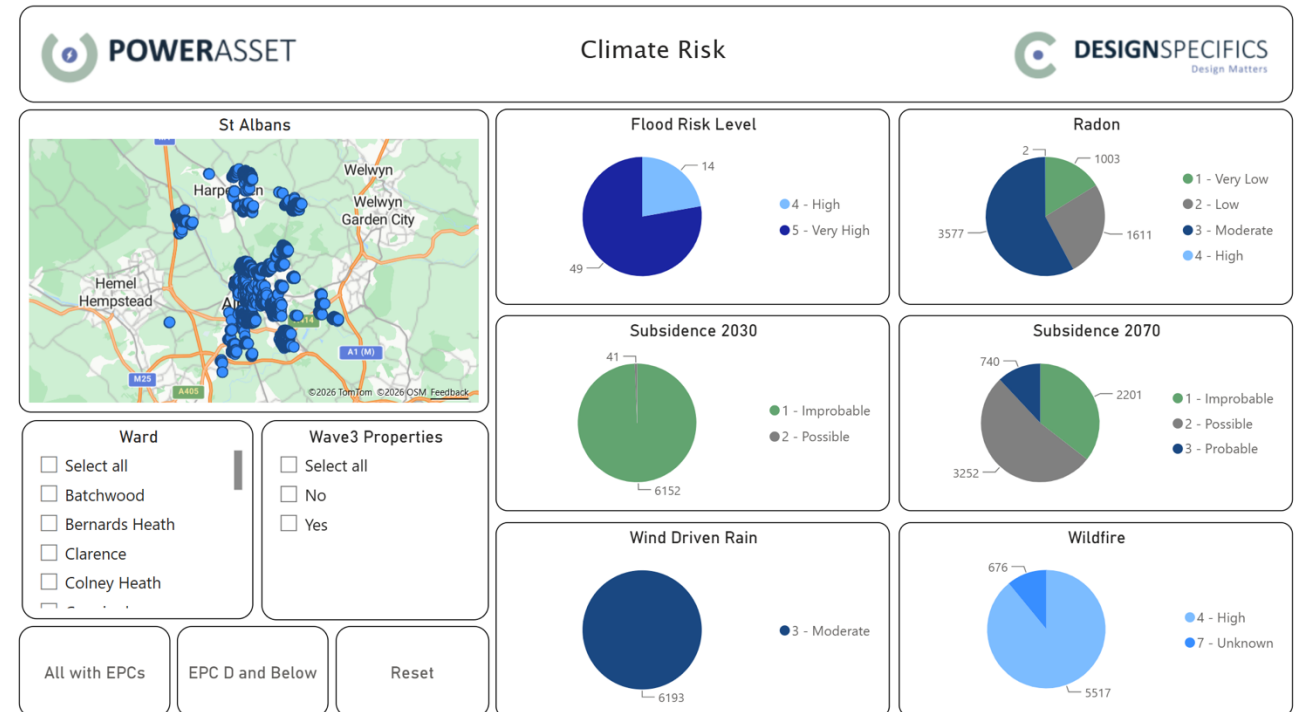
Climate Risk

PowerAsset has analysed climate related risks across the St Albans City & District Council portfolio to support long term investment planning and asset resilience.

Key findings include:

- **3,577 properties** are within areas of **moderate radon risk**, while **2 properties** are at **high risk**.
- **5,517 properties** are within areas with **high wildfire risk**.
- Over **60 properties** are located within **high or very high flood risk** areas.
- More than **6,000 properties** are affected by **moderate wind driven rain** exposure.

These insights enable ESG to identify where additional surveys, design considerations and mitigation measures may be required, ensuring investment programmes are resilient to future climate risks while reducing long term maintenance costs and improving the performance and durability of retrofit interventions.



Climate risks across St Albans City & District Council housing Portfolio

Geographical Clusters

PowerAsset has identified a number of deliverable geographical clusters that provide strong opportunities for area based retrofit delivery.

AL1 2L is one such example. The area contains a mix of **houses, bungalows, maisonettes and flats**, predominantly of **cavity wall** and **system-built** construction, making it well suited to archetype-based delivery. The cluster presents an excellent opportunity for a coordinated retrofit programme, reducing delivery complexity and improving economies of scale.

By combining IMD data with property-level housing, energy performance, tenure and eligibility data through PowerAsset, we can support WPS and St Albans City & District Council to develop an area-based, cross-tenure approach to retrofit. This enables programmes to be planned at street and neighbourhood level, focusing on areas with the strongest potential to deliver retrofit across multiple tenures and funding routes, including SHF Wave 3 and the Warm Homes: Local Grant. The analysis identifies eligible properties and tenure, groups homes into viable delivery clusters, and highlights opportunities to extend works to neighbouring properties, maximising funding, economies of scale and whole-street approach to retrofit.



Cluster of properties in AL1 2(x)

Digital Twins

Why We Use Digital Twins

DSL's use of digital twins is a core part of our retrofit approach and is delivered as standard at no additional cost to the client. This process is central to how we increase design accuracy, reduce delivery risk, and streamline project coordination across stakeholders.

Benefits of Digital Twins:

- **Enhanced precision:** LiDAR scanning and digital modelling ensure millimetre-level accuracy in retrofit design.
- **Improved efficiency:** Capturing everything digitally up front reduces rework and streamlines design coordination.
- **Lower cost:** Detailed models reduce the risk of design error, scope drift, and costly site changes - delivering savings over time.
- **Supporting Tools and Data Sources**

Our Digital Twin models are enriched through:

- **Drone Surveys:** Capture roof geometry and external façades with aerial precision—useful for solar PV feasibility, site logistics, and condition surveys.
- **Analysis:** Property and retrofit data is layered into GIS dashboards, helping target properties, manage delivery zones, and track live project metrics.
- **Thermography:** Thermal imaging is used alongside digital models to verify cold spots and validate retrofit impact.
- **Scan to BIM:** Our standard process to convert Point Cloud data into coordinated floor plans, elevations, and sections - used across all key archetypes.



LiDAR scanning enables precise, high-resolution capture of real-world environments; forming the foundation for accurate digital twins and informed planning.

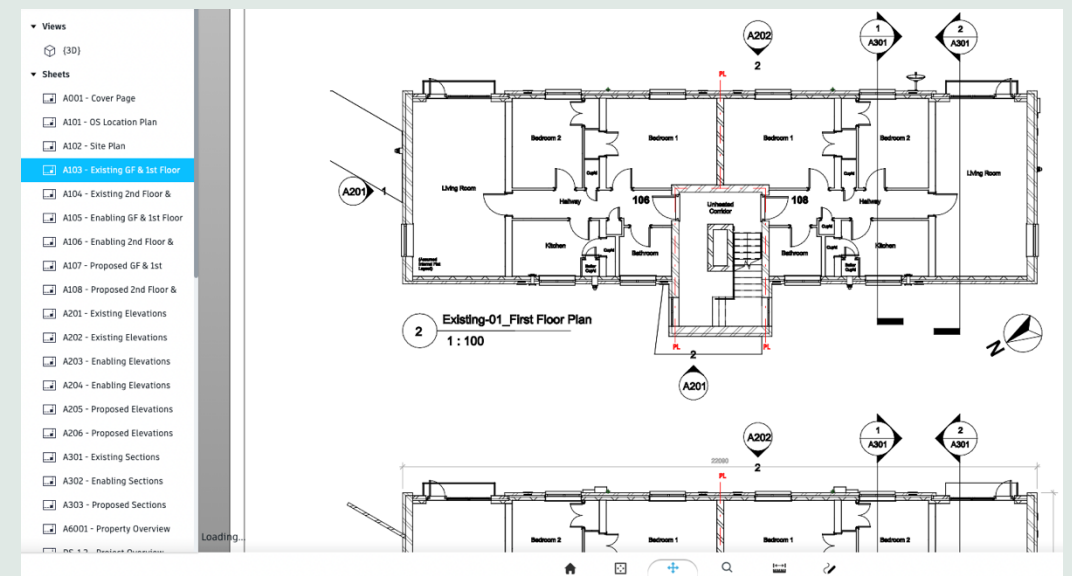
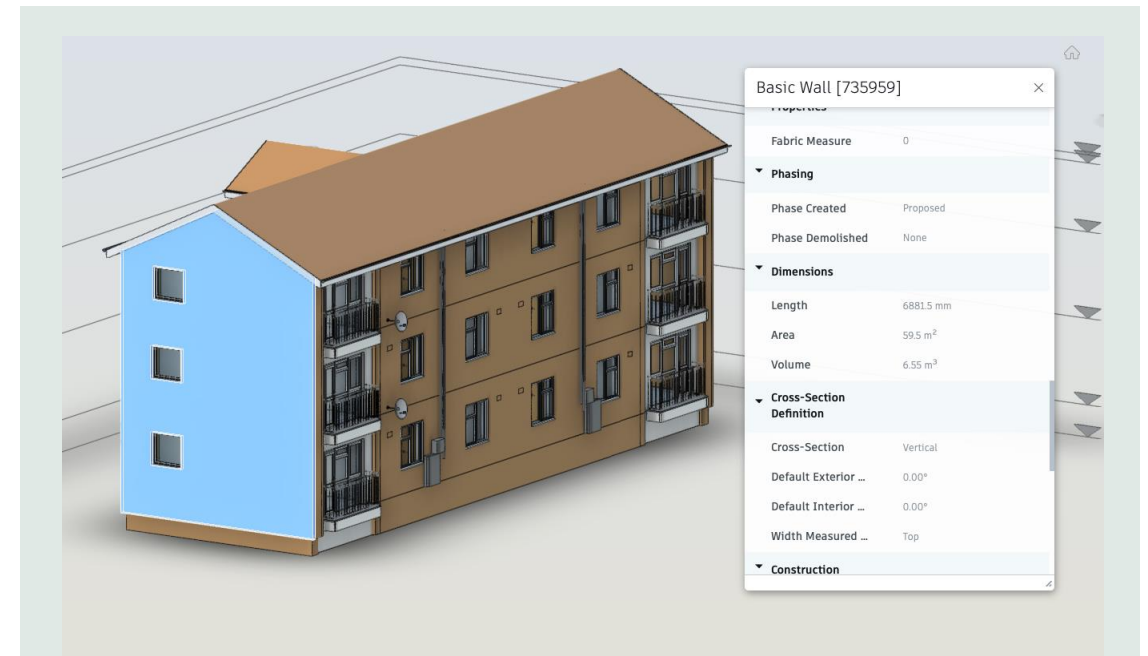


High-resolution point cloud scan capturing the precise geometry and spatial conditions of the site; a foundational step in creating an accurate digital twin.

Digital Twin Methodology - Use Cases

Proven Use Cases for Digital Twins:

- **Design Coordination & Compliance:**
 - Identify and address thermal bridges early (e.g. slab edges, window junctions)
 - Simulate overheating scenarios in dwellings with high solar gain or limited ventilation
 - Model optimal placement for **solar PV**, **ASHPs** & **radiators** from accurate thermal data
- **Reduce Void Turnaround time:**
 - With digital twins for each archetype ready with designs, site teams can be mobilised at speed, significantly reducing the void turnaround time
- **Tendering & Procurement:**
 - Generate reliable **Bill of Quantities (BoQs)** for tender packs, reducing cost uncertainty
 - Track product specifications used in design and ensure consistency across properties
- **Stakeholder Engagement:**
 - Liaise with **planners** using detailed models of existing conditions and proposed works
 - Provide **interactive 3D models** to brief installers and reduce misinterpretation of designs
 - Improve **tenant liaison** by visualising proposed works and helping build trust & understanding
- **Site Communication:**
 - Digital twins clarify issues remotely—minimising the need for repeated site visits and reducing downtime
 - On-site teams can access models through tablets or online viewers, ensuring fast, accurate interpretation
- **Future-Proofing:**
 - Serve as a digital record of the property's condition and works carried out
 - Allow future upgrades (e.g. battery storage, EV charging, second-phase insulation) to be modelled and planned efficiently



Example of a Digital Twin of a residential block of flats, which can include a wide array of data points such as the specification and product information of equipment within the homes or its installation date.

Conclusion

This high-level stock study provides St Albans City & District Council with a data-driven foundation for delivering an effective and scalable planned investment and retrofit programmes. By combining EPC, GIS, climate, asset, and archetype analysis, ESG has identified key retrofit opportunities, delivery challenges, and priority property groups across solid wall, cavity wall, and system-built homes.

The findings demonstrate the value of geographically focused retrofit schemes to improve logistics, reduce design variation, and maximise cost efficiency. Integrating complementary property types alongside existing delivery areas will support greater carbon savings, improved funding alignment, and more efficient mobilisation.

The analysis also highlights the importance of a whole-house and risk-based approach, considering factors such as non-traditional construction, overheating, conservation constraints, and planned maintenance opportunities. Further refinement using detailed ownership and estate data will help target investment where it can deliver the greatest impact.

ESG's work in collaboration with WPS, provides clear pathways for St Albans City & District Council to develop a practical, resident-focused, and future-ready retrofit programme, supporting decarbonisation goals while maximising value from investment.

